

# Adding and Subtracting in Scientific Notation

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## MA.8.NSO.1.5

I can add and subtract two numbers written in scientific notation, and write the answer in scientific notation.

# Warm-Up

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1. Write 0.00062 in scientific notation.
2. How many times larger is  $8 \times 10^7$  than  $2 \times 10^4$  ?
3. Find the value:  $\sqrt{144}$
4. Find the sum:  $0.49 + 1.2$

# Warm-Up

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1. Write 0.00062 in scientific notation.  **$6.2 \times 10^{-4}$**
2. How many times larger is  $8 \times 10^7$  than  $2 \times 10^4$  ? **4,000**
3. Find the value:  $\sqrt{144}$  **12**
4. Find the sum:  $0.49 + 1.2$  **1.69**

# Notes

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## I. A power of ten is a place value

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| $10^3$    | $10^4$        | $10^5$            | $10^6$   |
|-----------|---------------|-------------------|----------|
| thousands | ten thousands | hundred thousands | millions |

$$49,500 + 121,000 = 170,500$$

$$4.95 \times 10^4 + 12.1 \times 10^4 = 17.05 \times 10^4 = 1.705 \times 10^5$$

**Two numbers can be added only when they carry the same power of ten.**

# Notes

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## II. Same power: add or subtract the coefficients

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$$(5.2 \times 10^6) + (3.1 \times 10^6) = 8.3 \times 10^6$$

$$(7 \times 10^{-8}) + (6 \times 10^{-8}) = 13 \times 10^{-8} = 1.3 \times 10^{-7}$$

**Add or subtract the coefficients. The power of ten does not change.**

Then check the coefficient: if it is not between 1 and 10, the answer is not finished.

# Notes

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## III. Different powers: rewrite one of them first

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$$(9.4 \times 10^5) - (8.2 \times 10^3)$$

$$= (9.4 \times 10^5) - (0.082 \times 10^5) = 9.318 \times 10^5$$

$$(1.3 \times 10^3) + (3.4 \times 10^5) = 3.413 \times 10^5, \quad \text{never} \quad 4.7 \times 10^8$$

**Rewrite one number so both carry the same power of ten, then add or subtract the coefficients.**

Exponent up one, decimal point one place left. The value does not move.

# Example 1

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$$(8.6 \times 10^7) + (5.9 \times 10^7)$$

**Add. Write the answer in scientific notation.**

# Example 1

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$$(8.6 \times 10^7) + (5.9 \times 10^7) = 14.5 \times 10^7$$

$$14.5 \times 10^7 = 1.45 \times 10^8$$

145,000,000

**Answer:  $1.45 \times 10^8$**

# Your Turn

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$$(4.8 \times 10^{-3}) + (7.5 \times 10^{-3})$$

**Add. Write the answer in scientific notation.**

# Your Turn

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$$(4.8 \times 10^{-3}) + (7.5 \times 10^{-3})$$

**Add. Write the answer in scientific notation.**

Same power, so add the fronts:  $4.8 + 7.5 = 12.3$

$$12.3 \times 10^{-3} = 1.23 \times 10^{-2}$$

**Answer:  $1.23 \times 10^{-2}$**

## Example 2

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$$(7.3 \times 10^6) - (4.5 \times 10^5)$$

**Subtract. Write the answer in scientific notation.**

## Example 2

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$$4.5 \times 10^5 = 0.45 \times 10^6$$

$$(7.3 \times 10^6) - (0.45 \times 10^6) = 6.85 \times 10^6$$

$$7,300,000 - 450,000 = 6,850,000$$

**Answer:  $6.85 \times 10^6$**

# Your Turn

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$$(5.6 \times 10^{-4}) - (9.0 \times 10^{-6})$$

**Subtract. Write the answer in scientific notation.**

# Your Turn

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$$(5.6 \times 10^{-4}) - (9.0 \times 10^{-6})$$

**Subtract. Write the answer in scientific notation.**

Match the powers:  $9.0 \times 10^{-6} = 0.09 \times 10^{-4}$

$$5.6 - 0.09 = 5.51$$

$$5.51 \times 10^{-4}$$

**Answer:  $5.51 \times 10^{-4}$**

# Whiteboards • Question 1 of 9

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$$(3.2 \times 10^5) + (4.1 \times 10^5)$$

**Answer it.**

# Whiteboards · Question 1 of 9

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$$(3.2 \times 10^5) + (4.1 \times 10^5)$$

Same power, so add the fronts:  $3.2 + 4.1 = 7.3$

$$7.3 \times 10^5$$

**Answer:  $7.3 \times 10^5$**

# Whiteboards • Question 2 of 9

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A stadium sold  $4.2 \times 10^4$  tickets on Friday  
and  $3.5 \times 10^4$  on Saturday. Each ticket cost 38 dollars.

**How many tickets were sold on the two days?**

**Answer it.**

# Whiteboards • Question 2 of 9

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A stadium sold  $4.2 \times 10^4$  tickets on Friday  
and  $3.5 \times 10^4$  on Saturday. Each ticket cost 38 dollars.

**How many tickets were sold on the two days?**

Same power, so add the fronts:  $4.2 + 3.5 = 7.7$   
 $7.7 \times 10^4$

**Answer:  $7.7 \times 10^4$**

# Whiteboards • Question 3 of 9

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$$(6 \times 10^{-5}) + (7 \times 10^{-5})$$

**Answer it.**

# Whiteboards · Question 3 of 9

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$$(6 \times 10^{-5}) + (7 \times 10^{-5})$$

$$6 + 7 = 13$$

$$13 \times 10^{-5} = 1.3 \times 10^{-4}$$

**Answer:  $1.3 \times 10^{-4}$**

# Whiteboards · Question 4 of 9

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$$(5.6 \times 10^8) - (4.9 \times 10^8)$$

**Answer it.**

# Whiteboards · Question 4 of 9

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$$(5.6 \times 10^8) - (4.9 \times 10^8)$$

$$5.6 - 4.9 = 0.7$$

$$0.7 \times 10^8 = 7 \times 10^7$$

**Answer:  $7 \times 10^7$**

# Whiteboards • Question 5 of 9

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$$(1.3 \times 10^3) + (3.4 \times 10^5)$$

Which is the sum?

**A.**  $4.7 \times 10^5$

**B.**  $4.7 \times 10^8$

**C.**  $3.413 \times 10^5$

**D.**  $3.53 \times 10^5$

**Answer it.**

# Whiteboards • Question 5 of 9

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$$(1.3 \times 10^3) + (3.4 \times 10^5)$$

Which is the sum?

Match the powers:  $1.3 \times 10^3 = 0.013 \times 10^5$

$$0.013 + 3.4 = 3.413$$

$$3.413 \times 10^5$$

**Answer: C —  $3.413 \times 10^5$**

# Whiteboards • Question 6 of 9

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$$(2.5 \times 10^6) + (9.8 \times 10^6)$$

Which is the sum, written in scientific notation?

**A.**  $12.3 \times 10^6$

**B.**  $12.3 \times 10^{12}$

**C.**  $1.23 \times 10^6$

**D.**  $1.23 \times 10^7$

**Answer it.**

# Whiteboards • Question 6 of 9

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$$(2.5 \times 10^6) + (9.8 \times 10^6)$$

Which is the sum, written in scientific notation?

$$2.5 + 9.8 = 12.3$$

$$12.3 \times 10^6 = 1.23 \times 10^7$$

**Answer: D —  $1.23 \times 10^7$**

# Whiteboards • Question 7 of 9

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$$(7.8 \times 10^{-2}) - (6.4 \times 10^{-3})$$

**Answer it.**

# Whiteboards • Question 7 of 9

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$$(7.8 \times 10^{-2}) - (6.4 \times 10^{-3})$$

Match the powers:  $6.4 \times 10^{-3} = 0.64 \times 10^{-2}$

$$7.8 - 0.64 = 7.16$$

$$7.16 \times 10^{-2}$$

**Answer:  $7.16 \times 10^{-2}$**

# Whiteboards • Question 8 of 9

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$$(4.5 \times 10^6) + (n \times 10^6) = 1.02 \times 10^7$$

What is the value of  $n$  ?

**Answer it.**

# Whiteboards · Question 8 of 9

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$$(4.5 \times 10^6) + (n \times 10^6) = 1.02 \times 10^7$$

What is the value of  $n$  ?

$$1.02 \times 10^7 = 10.2 \times 10^6$$

$$4.5 + n = 10.2$$

$$n = 10.2 - 4.5 = 5.7$$

**Answer:  $n = 5.7$**

# Whiteboards • Question 9 of 9

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Devon writes  $(1.3 \times 10^3) + (3.4 \times 10^5) = 4.7 \times 10^8$ .

Name the two mistakes he made, and give the correct sum.

**Write your answer in sentences.**

# Whiteboards • Question 9 of 9

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Devon writes  $(1.3 \times 10^3) + (3.4 \times 10^5) = 4.7 \times 10^8$ .

Name the two mistakes he made, and give the correct sum.

The powers do not match, and exponents are not added.

$$1.3 \times 10^3 = 0.013 \times 10^5$$

$$0.013 + 3.4 = 3.413, \text{ so } 3.413 \times 10^5$$

**Answer:  $3.413 \times 10^5$**

# Independent Set

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1.  $(4.1 \times 10^6) + (2.7 \times 10^6)$
2.  $(9.3 \times 10^{-5}) - (5.1 \times 10^{-5})$
3.  $(7.5 \times 10^3) + (8.9 \times 10^3)$
4.  $(6.2 \times 10^7) + (3.5 \times 10^6)$
5.  $(4.8 \times 10^{-2}) - (9.0 \times 10^{-4})$
6. Find  $n$ .  $(5.2 \times 10^9) + (n \times 10^9) = 1.31 \times 10^{10}$

# IXL

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1. Open IXL and start today's skills.
2. Work on paper where the question needs work. The answer box does not show it.
3. Every skill listed is required, to a SmartScore of 67. Due at the start of the next class.

## **Today's skills — all required**

Add and subtract numbers written in scientific notation — HUR